Model Formulations and Development

1. Research Question

Which cities should the company enter, and how many vehicles should it allocate to each city? The underlying document develops three formulations: a basic model, a cost-minimization model, and a profit-maximization model. City entry decisions y_c and vehicle allocations q_c are determined jointly before the demand scenario is realized. The same allocation applies across all scenarios.

2. Core Structure

Decisions	$y_c \in \{0,1\}$: city entry; $q_c \in \mathbb{Z}_{\geq 0}$: number of vehicles
Inputs	D_c^s: scenario-specific demand; κ_c: service capacity per vehicle; Q: total fleet limit; $\bar{q}_c$: city-specific vehicle limit
Outcomes	$R_c^s = \min(D_c^s, \kappa_c q_c)$: served demand $O_c^s = (\kappa_c q_c - D_c^s)^+$: unused service capacity $U_c^s = (D_c^s - \kappa_c q_c)^+$: unmet demand

Here, $(x)^+ = \max(x, 0)$. Once generated, demand scenarios are fixed inputs; demand is not chosen by the optimization model.

3. Comparison of the Three Objectives

Model	Objective and interpretation
1. Basic	$\min\ \mathbb{E}_s\left[\sum_c (O_c^s + \rho U_c^s)\right]$ Balances unused capacity and unmet demand. The parameter ρ measures the relative importance of shortages. Entry and vehicle costs are excluded.
2. Cost	$\min\ \sum_c (F_c y_c + K_c q_c) + \mathbb{E}_s\left[\sum_c (c_c^+ O_c^s + c_c^- U_c^s)\right]$ Minimizes entry costs, vehicle costs, unused-capacity penalties, and unmet-demand losses. **This is the current baseline objective.**
3. Profit	$\max\ \mathbb{E}_s\left[\sum_c m_c R_c^s\right] - \sum_c (F_c y_c + K_c q_c)$ Deducts entry and vehicle costs from the contribution margin on served trips. Here, m_c is revenue per trip minus its variable cost.

4. Common Constraints and Subsequent Revisions

$$y_c \leq q_c \leq \bar{q}_c y_c, \qquad \sum_c q_c \leq Q, \qquad \sum_c y_c \leq B, \qquad y_c \leq a_c.$$

Here, a_c indicates whether operation in city c is feasible. The parameter B limits the number of cities that can be entered simultaneously and is used only when such a restriction actually exists.

5. Key Interpretation Notes

- **Avoid double counting.** Do not include costs already covered by K_c again in c_c^+. In the profit model, do not deduct the same forgone contribution margin again as an unmet-demand loss.
- **Relationship between models.** Under identical demand and constraints, if $c_c^+ = 0$ and $c_c^- = m_c$, the cost and profit models yield the same optimal allocations. The cost objective equals a demand-dependent constant minus the profit objective.
- **Unused capacity versus its penalty.** Unused capacity O_c^s can be positive even when $c_c^+ = 0$. Only the additional unused-capacity penalty is zero; vehicle costs remain.
- **Model development.** A fourth, multi-penalty formulation was introduced in the original outline. Following feedback on that formulation, Model 2 (cost minimization) was adopted as the baseline.